Do Now · retrieval

Complexity and algorithm choice

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Why is a queue suitable for a print server?

Q2

What condition must the array satisfy?

Q3

How many times does a for-loop from i=0 to i<6 execute?

Q4

A loop of length 4 inside another loop of length 4: how many inner operations?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Start with n — n is the number of items. More items usually mean more operations.

Core idea 2

Trace loop nesting — One full scan performs n steps. A nested n-by-n scan performs n^2 steps.

Core idea 3

Compare growth — When n doubles, linear work doubles; quadratic work roughly quadruples. Big O describes growth, not seconds.

Key visual · complete concept

Complexity and algorithm choice

n = 8	Linear 8	Quadratic 64
n = 16	Linear 16	Quadratic 256
n = 32	Linear 32	Quadratic 1024

Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

Big O — Asymptotic growth of algorithm work.

O(1) — Constant growth.

O(log n) — Logarithmic growth, e.g. halving search intervals.

O(n) — Linear growth.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

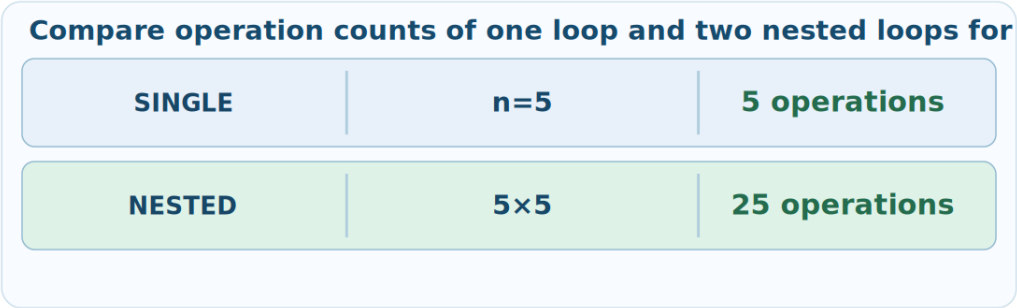
Problem

Compare operation counts of one loop and two nested loops for n=5.

Follow the worked method

1. Predict — One loop visits five elements, so does five operations.
2. Explain — The nested loop repeats five operations five times, making 25.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict

Linear search can check all 64 values.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Give the Big O of a single full scan of n elements.

Question 2

For n=100, give iterations of a linear n loop and nested n×n loop.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Why may linear search be preferable for a single search of a short unsorted array?

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

$O(n^2)$ means an algorithm takes n^2 seconds.

Use key vocabulary from memory

1. Explain Big O in your own words.

2. Explain O(1) in your own words.

3. Explain O(log n) in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.